

Enhanced Software Requirements Specification for Timetable Management System

Version 2.0

22 August 2025

Prepared for:

Asanska University College of Design and Technology

Enhanced with IEEE Standards Compliance
Accessibility Excellence & Brand Integration

Contents

1. Executive Summary

This enhanced Software Requirements Specification (SRS) document incorporates IEEE standards compliance, comprehensive accessibility requirements, and full integration with AUCDT brand identity guidelines. This revision addresses critical gaps in the original specification while maintaining the core functional objectives.

Critical Requirement

This document supersedes version 1.3 and includes mandatory accessibility compliance requirements, enhanced security specifications, and detailed brand implementation guidelines that are legally required for institutional software systems.

2. Introduction

2.1 Purpose

This Software Requirements Specification (SRS) document defines the comprehensive functional and non-functional requirements for the Timetable Management System (TMS) for the Asanska University College of Design and Technology. This enhanced specification ensures IEEE 830-1998 compliance, Web Content Accessibility Guidelines (WCAG) 2.1 Level AA conformance, and full integration with AUCDT brand identity standards.

2.2 Document Conventions

This document adheres to:

- IEEE Std 830-1998 recommended practices for Software Requirements Specifications
- Section 508 accessibility compliance standards
- AUCDT Brand Identity Guide v2.0 specifications
- ISO/IEC 25010:2011 software quality model

All requirements include unique identifiers, priority classifications, and measurable acceptance criteria.

2.3 Intended Audience and Reading Suggestions

Primary Audience:

- Project managers and technical architects
- Software developers and QA engineers

- Accessibility compliance officers
- University administrators and end users
- Legal and compliance teams

Reading Sequence: Review Executive Summary ☒ Overall Description ☒ Accessibility Requirements ☒ System Features ☒ Implementation Constraints.

2.4 Enhanced Product Scope

The TMS encompasses a comprehensive, accessible, brand-compliant web-based ecosystem including:

- Multi-modal user interfaces (visual, auditory, tactile)
- Real-time collaboration and synchronization
- Advanced analytics and predictive scheduling
- Comprehensive API ecosystem for integrations
- Mobile-first responsive design with offline capabilities
- Multi-language support architecture

3. Enhanced System Features

3.1 Universal Access Authentication

Authentication & Authorization

Priority: Critical

Compliance: Section 508, OAuth 2.0, SAML 2.0

Functional Requirements:

REQ-001 Multi-factor authentication supporting hardware tokens, biometric authentication, and assistive technology compatibility

REQ-002 Single Sign-On (SSO) integration with university identity providers

REQ-003 Role-based access control with granular permissions and audit trails

REQ-004 Session management with configurable timeout and concurrent session limits

REQ-005 Password-less authentication options for users with cognitive disabilities

3.2 Intelligent Timetable Engine

Advanced Scheduling System

Priority: Critical

Performance Target: <30 seconds for complex schedules

Functional Requirements:

REQ-006 AI-powered conflict resolution using constraint satisfaction algorithms

REQ-007 Real-time collaborative editing with operational transformation

REQ-008 Template-based scheduling with version control and rollback capabilities

REQ-009 Multi-criteria optimization considering accessibility requirements, travel distances, and resource utilization

REQ-010 Scenario modeling and "what-if" analysis tools

3.3 Comprehensive Accessibility Features

⚠ Critical Requirement

Legal Requirement: Full WCAG 2.1 Level AA compliance is mandatory for educational institutions.

Accessibility Requirements:

REQ-011 Visual Accessibility: High contrast modes, customizable font sizes (up to 200% zoom), alternative color schemes for color-blind users

REQ-012 Motor Accessibility: Full keyboard navigation, customizable keyboard shortcuts, voice control integration, switch control support

REQ-013 Cognitive Accessibility: Simplified interface modes, consistent navigation patterns, clear error messages, extended session timeouts

REQ-014 Auditory Accessibility: Visual indicators for all audio cues, captions for video content, text-to-speech compatibility

REQ-015 Assistive Technology: Screen reader optimization (NVDA, JAWS, VoiceOver), voice recognition software compatibility

3.4 Brand-Compliant User Interface

Brand Compliance

AUCDT Brand Integration Requirements:

REQ-016 Color Palette: Mandatory use of AUCDT brand colors - Burgundy Primary (#8B1538), Gold Accent (#D4AF37), Cream Background (#F8F6F0)

REQ-017 Typography: Poppins font family implementation with proper hierarchy (Bold for H1, SemiBold for H2, Medium for H3, Regular for body)

REQ-018 UI Components: Brand-compliant buttons, cards, and containers matching AUCDT Brand Identity Guide v2.0

REQ-019 Logo Usage: Proper logo implementation following brand protection guidelines with minimum clear space requirements

REQ-020 Accessibility Compliance: Ensure all brand colors meet WCAG contrast ratios (4.5:1 for normal text, 3:1 for large text)

4. Enhanced Non-Functional Requirements

4.1 Performance Requirements

Table 1: Performance Benchmarks

Operation	Target Time	Maximum Time
Page Load (Initial)	<2 seconds	3 seconds
Page Load (Subsequent)	<1 second	2 seconds
Search Results	<500ms	1 second
Timetable Generation	<30 seconds	45 seconds
Real-time Updates	<2 seconds	3 seconds
Mobile Response	<3 seconds	4 seconds
Offline Sync	<10 seconds	15 seconds

4.2 Enhanced Security Requirements

Comprehensive Security Framework

- REQ-021 Data Protection:** AES-256 encryption for data at rest, TLS 1.3 for data in transit
- REQ-022 API Security:** OAuth 2.0 with PKCE, rate limiting (100 req/min/user), API key management
- REQ-023 Audit Logging:** Immutable, cryptographically signed audit trails with blockchain verification
- REQ-024 Privacy Compliance:** GDPR, FERPA compliance with data retention policies and user consent management
- REQ-025 Vulnerability Management:** Regular security scans, penetration testing, dependency vulnerability monitoring

4.3 Internationalization Requirements

Multi-Language Support

- REQ-026 Language Support:** English (primary), French, Arabic with RTL support
- REQ-027 Localization:** Currency, date/time formats, number formats, cultural preferences
- REQ-028 Content Management:** Translation management system with version control
- REQ-029 Accessibility:** Screen reader support in all supported languages

5. Integration Requirements

5.1 Educational Technology Standards

Standards Compliance

REQ-030 LTI Compliance: Learning Tools Interoperability (LTI) 1.3 implementation

REQ-031 QTI Support: Question and Test Interoperability for assessment integration

REQ-032 xAPI Integration: Experience API for learning analytics

REQ-033 OneRoster: Student information system data exchange

5.2 Third-Party Integrations

Table 2: Required Integrations

Service	Purpose	Priority
Microsoft 365	Calendar, Email	High
Google Workspace	Calendar, Drive	High
Moodle LMS	Course Integration	Critical
Campus Card System	Access Control	Medium
Library System	Resource Booking	Low

6. Quality Assurance Requirements

6.1 Testing Requirements

⚠ Critical Requirement

Mandatory Testing Procedures:

- Automated accessibility testing using axe-core and Pa11y
- Manual accessibility testing with disabled users
- Cross-browser testing including assistive technologies
- Performance testing under realistic load conditions
- Security penetration testing by certified ethical hackers
- Usability testing with representative user groups

6.2 Acceptance Criteria

System Acceptance Requirements:

- 100% WCAG 2.1 Level AA compliance verification
- 99.5% uptime during 30-day evaluation period
- User satisfaction score >85% across all user groups
- Brand compliance audit by AUCDT Marketing Department
- Security audit clearance by institutional IT security team

7. Implementation Roadmap

7.1 Phased Delivery Approach

Phase 1 (Months 1-3): Core authentication, basic timetable viewing, accessibility framework

Phase 2 (Months 4-6): Advanced scheduling, notification system, mobile optimization

Phase 3 (Months 7-9): Analytics, reporting, third-party integrations

Phase 4 (Months 10-12): Multi-language support, advanced accessibility features, AI optimization

7.2 Risk Mitigation

Risk Management Strategy

RISK-001 Accessibility Compliance: Regular audits, disabled user testing, legal consultation

RISK-002 Performance: Load testing, CDN implementation, caching strategies

RISK-003 Data Migration: Comprehensive backup, rollback procedures, parallel systems

RISK-004 User Adoption: Training programs, change management, user support

8. Conclusion

This enhanced SRS establishes a comprehensive framework for developing a world-class, accessible, and brand-compliant Timetable Management System. The requirements ensure legal compliance, technical excellence, and exceptional user experience for all members of the AUCDT community.

Next Steps:

1. Stakeholder review and approval

2. Technical architecture design
3. Accessibility consultant engagement
4. Development team onboarding
5. Pilot user group establishment

Document Approval Required: IT Director, Accessibility Officer, Legal Counsel,
Academic Registrar

Next Review Date: January 2026

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